
DSC 140A - Homework 03

Due: Wednesday, January 29

Instructions: Write your solutions to the following problems either by typing them or handwriting them on another piece of paper or on an iPad/tablet. Show your work or provide justification unless otherwise noted; submissions that don't show work might lose credit. If you write code to solve a problem, include the code by copy/pasting or as a screenshot. You may use `numpy`, `pandas`, `matplotlib` (or another plotting library), and any standard library module, but no other third-party libraries unless specified. Submit homeworks via Gradescope by 11:59 PM.

A L^AT_EX template is provided at <http://dsc140a.com>, next to where you found this homework. Using it is totally optional, but encouraged if you plan to go to grad school. See this video for a quick introduction to L^AT_EX.

Problem 1.

Suppose that $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is convex and $g : \mathbb{R} \rightarrow \mathbb{R}$ is convex and non-decreasing. That is, if $a > b$, then $g(a) \geq g(b)$.

Show that the composition of these functions, $h(\vec{x}) = g(f(\vec{x}))$, is also convex.

Hint: you'll want to go back to the definition to show this is true. A similar problem was solved in discussion.

Problem 2.

The *hinge loss* is defined to be

$$L_{\text{hinge}}(\vec{w}, \vec{x}, y) = \max\{0, 1 - y \vec{w} \cdot \text{Aug}(\vec{x})\}$$

Consider the below quantity, which is called the *regularized* empirical risk:

$$R(\vec{w}) = \frac{C}{n} \sum_{i=1}^n L_{\text{hinge}}(\vec{w}, \vec{x}^{(i)}, y_i) + \|\vec{w}\|^2$$

Here, C is a positive constant, n is the number of data points, $\vec{x}^{(i)}$ is the i th data point, y_i is the label of the i th data point, and $\vec{w}$ is a vector of weights.

Show that $R(\vec{w})$ is a convex function of $\vec{w}$.

(We will discuss regularization and the hinge loss in future lectures; for this problem it's not necessary to know anything about them apart from the definitions given above.)

Hint: you will probably *not* want to use the formal definition of convexity here. Instead, you'll want to show that R is composed of simpler functions which themselves are convex.

Problem 3.

Consider a linear prediction function H used for binary classification, and assume that when the output of H is positive we predict for class +1, and when it's negative we predict for class -1. This means that the **decision boundary** is where $H(\vec{x}) = 0$.

In lecture, we saw that a linear prediction function has the form:

$$\begin{aligned} H(\vec{x}) &= w_0 + w_1 x_1 + \dots + w_d x_d \\ &= \vec{w} \cdot \text{Aug}(\vec{x}) \end{aligned}$$

where $\vec{w} = (w_0, w_1, \dots, w_d)^T$. In this problem, it will also be useful to define the vector $\vec{w}' = (w_1, \dots, w_d)^T$, which is the same as $\vec{w}$ except that it does not include w_0 . Note that $\vec{x} \cdot \vec{w}' = w_1 x_1 + \dots + w_d x_d$. With this definition, we can write $H(\vec{x})$ in a slightly different way:

$$H(\vec{x}) = w_0 + \vec{w}' \cdot \vec{x}$$

Remember this formula, as it will be useful several times below!

Over the course of this problem, we'll answer the question: how is the magnitude of $H(\vec{x})$ related to the distance between $\vec{x}$ and the decision boundary?

Note: for this problem it may be useful to review the properties of the dot product and vector algebra. In particular, remember that $\|u\|$ denotes the norm (length) of a vector, and that $\vec{u} \cdot \vec{u} = \|\vec{u}\|^2$. By dividing a vector by its norm, as in $\vec{u}/\|\vec{u}\|$, we obtain a *unit vector* with unit length – a unit vector is useful for specifying a direction. To find the component of a vector $\vec{u}$ that points in the same direction as another vector $\vec{v}$, we write $\vec{u} \cdot \vec{v} / \|\vec{v}\|$. Also remember that $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$, and that $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$.

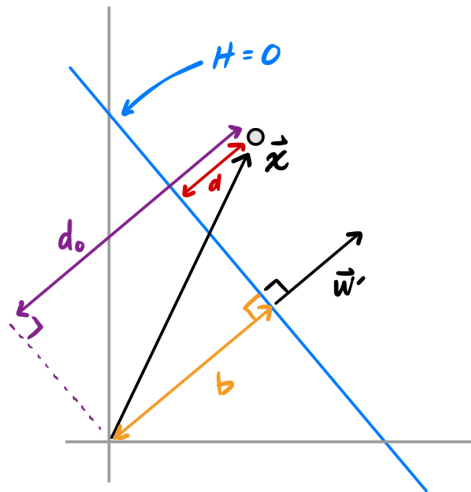
- a) Show that for any point $\vec{z}$ on the decision boundary, $\vec{w}' \cdot \vec{z} = -w_0$.

Hint: use that fact that $H(\vec{z}) = 0$.

- b) Argue that $\vec{w}'$ is orthogonal to the decision boundary.

Hint: take two arbitrary points $\vec{x}^{(1)}$ and $\vec{x}^{(2)}$ that are assumed to be on the decision boundary. Then, since we know that the boundary is linear (it is a line, plane, etc.), the difference of these vectors, $\vec{\delta} = \vec{x}^{(1)} - \vec{x}^{(2)}$ is parallel to the boundary. To show that $\vec{w}'$ is orthogonal to the boundary, it suffices to show that $\vec{w}' \cdot \vec{\delta} = 0$. Make sure you somehow use the fact that $\vec{x}^{(1)}$ and $\vec{x}^{(2)}$ are on the decision boundary.

- c) Now that we know that $\vec{w}'$ is orthogonal to the decision boundary, we can draw a better picture of the situation:



The blue line is the decision boundary – it is where $H = 0$. We have drawn an arbitrary point $\vec{x}$, along with several distances:

- d : the (signed) distance from the decision boundary to $\vec{x}$
- b : the distance from the the origin to the decision boundary

- d_0 : the length of the component of $\vec{x}$ that is orthogonal to the decision boundary.

We're most interested in knowing d . First, though, we need to find b .

Consider the vector $b\vec{w}'/\|\vec{w}'\|$; this is a vector from the origin to the decision boundary that is orthogonal to the boundary and with length b .

Since this vector is on the decision boundary, $H(b\vec{w}'/\|\vec{w}'\|) = 0$. Using this fact, show that $b = -\frac{w_0}{\|\vec{w}'\|}$.

Hint: first show that $H(b\vec{w}'/\|\vec{w}'\|) = b\|\vec{w}'\| + w_0$, then set this to zero and solve for b .

- d) Recall that d_0 is the component of $\vec{x}$ that is orthogonal to the decision boundary; this is simply $\vec{x} \cdot \vec{w}'/\|\vec{w}'\|$. From the picture, $d_0 = d + b$. We know d_0 and b , and can therefore solve for d .

Use this to show that $|d| = |H(\vec{x})|/\|\vec{w}'\|$.

We have shown that the distance between $\vec{x}$ and the decision boundary is proportional to the output of the prediction function, $H(\vec{x})$. This gives us a very useful interpretation of $|H(\vec{x})|$! For example, this means if $H(\vec{x}^{(1)}) > H(\vec{x}^{(2)}) > 0$, then $\vec{x}^{(1)}$ is further from the decision boundary than $\vec{x}^{(2)}$.

Problem 4.

The file below contains data suitable for a binary classification problem. https://f000.backblazeb2.com/file/jeldridge-data/003-two_clusters/data.csv

The file contains three columns: x_1 , x_2 , and y . The first two columns are the features, and the third column is the label.

- Write a function to compute the empirical risk for the perceptron loss, $R_{\text{tron}}(\vec{w})$, with respect to the data and the parameters $\vec{w}$.
Use your function to compute the empirical risk for the vector $(1, 1, 1)^T$, and report the result. Include your code.
- Write a function to compute the subgradient of the empirical risk for the perceptron loss, and use your code to compute the subgradient at the point $(1, 1, 1)^T$. Report the result.
- Run subgradient descent to train a perceptron on the data. Use the initial vector $(1, 1, 1)^T$. Report the learned parameter vector.
- Plot the data and the decision boundary of the perceptron you learned above. Each point should be colored according to its label.

Hint: There are two ways to plot the decision boundary. One way is to use the fact that $\text{Aug}(\vec{x}) \cdot \vec{w} = 0$ for points on the decision boundary to solve for x_2 in terms of x_1 , giving you the equation of a line. Another approach is to use `plt.contour` to plot the contours of the prediction function, $H(\vec{x})$; in particular, we plot the contour where $H(\vec{x}) = 0$. The second approach is more general and can be used for any prediction function, but you'll want to read the docs to understand how to use it.